Shivansh Tyagi

LinkedIn | shivelite2003@gmail.com | +91-7456833022 | Github | Portfolio

ML/AI Engineer specializing in **PyTorch, Deep Learning**, and high-performance data pipelines. Expert in building persistent memory systems for LLM agents. Recognized for technical excellence as a **Smart India Hackathon Finalist** and **first-author of an IEEE publication.**

Education

Raj Kumar Goel Institute of Technology (AKTU) — **2022-2026**
*Bachelor of Technology in Computer Science and Engineering (**CGPA: 7.7**)* — *Ghaziabad, Uttar Pradesh*

Skills

ML & AI: PyTorch, ONNX Runtime, Transformers, LangChain, RAG, Vector DBs (Qdrant, ChromaDB)
Geospatial: GDAL, GeoPandas, Shapely, QGIS, Fiona
Frameworks: FastAPI, Flask, Streamlit
Databases: PostgreSQL, MongoDB, Redis
DevOps: Docker, Kubernetes, Git, AWS
Programming Languages: Python

Experience

Government of India — **Jan 2026 - Present**
Machine Learning Intern

- Re-engineered a fragmented two-script imagery workflow into a single automated end-to-end pipeline using a Watchdog-driven **producer-consumer** architecture, **reducing 70GB+ daily PLANET imagery processing from 4 hours to 40 minutes (6× speedup)**
- Accelerated TIFF preprocessing via GPU-based percentile normalization (**PyTorch** tensors), tiled **GDAL** ReadRaster to avoid full-scene RAM loading, **tmpfs** RAM-disk staging, and batched **ONNX** Runtime inference for classification
- Developed the evaluation scripts for a **national-level startup challenge**, benchmarking ML models for vessel path prediction and AIS correlation against proprietary ground-truth datasets.
- Migrating legacy AIS decoders to a fault-tolerant modern architecture, targeting reduced error rates and improved throughput

Publications

Cognitively-Inspired Dual Memory for Long-Term Contextual Reasoning — **IEEE, GTSS 2026**
First Author *— Presented — In Press*

- Proposed D-mem, a dual-memory architecture separating factual (Semantic) and experiential (Episodic) knowledge via a bi-temporal knowledge graph (**Neo4j**) to enable **"as-of"** and **"since-when"** contextual reasoning.

Projects

EchoDB — Temporal Memory System for LLM Agents | *github* | *pypi*

- **Architected and published** a persistent memory layer for LLM agents that enables long-term fact retention, structured knowledge extraction, and automated entity linking across conversation sessions.
- Engineered a **temporal graph framework** using **versioned chains** to manage state changes over time, ensuring agents retrieve the latest information while preserving a traceable historical context for updated facts.
- Implemented a **biologically-inspired memory** decay system and hybrid retrieval algorithm that prioritizes relevant context by balancing semantic similarity with access frequency and recency.
- Developed a **portable snapshot system (.umc)** storing graph + embeddings for instant restore without reprocessing

CLIP-Guided Image Optimization Engine - Dreams | *github*

- Engineered a custom **PyTorch pipeline** using **OpenCLIP (ViT-B/32)** to perform gradient ascent on image tensors, implementing custom differentiable **loss functions (Total Variation, Range Loss)** to maximize semantic alignment while preserving structural edge integrity.
- Designed a memory-efficient **multi-octave** upscaling algorithm with gradient smoothing and batched cutout augmentation, enabling high-fidelity image generation on consumer-grade GPUs with limited VRAM.

Achievements

- **Smart India Hackathon 2025 Finalist — Led 6-member team; one of 2 teams to deliver a working product through the final round of India's largest national govt. hackathon**
- Winning the prize for Best use of MongoDb Atlas in HackHound 3.0.
- Achieved rank **8** out of 300+ teams in **IIT Delhi's Tryst'25**
- Secured **1st** place in college Code Clash event **3 times**, awarded by Striver.
- Rated **3** star at Codechef, **1127** on Codeforces.